Development for an Explainable and Reliable Artificial Intelligence Framework for Preeclampsia Prediction and Pregnancy Risk Assessment

Engineering / Master's Project

École Supérieure en Informatique – Sidi Bel Abbès (ESI-SBA)

Research Domain: Artificial Intelligence – Machine Learning – Explainable AI
Biomedical Informatics – Clinical Decision Support

1 Project Context

Preeclampsia is a major pregnancy complication associated with maternal and fetal risks. Early identification for women at increased risk can support closer monitoring and appropriate clinical management.

Recent advances in Artificial Intelligence (AI) and Machine Learning (ML) provide opportunities to exploit heterogeneous clinical information for pregnancy-risk prediction. Maternal characteristics, uterine artery Doppler measurements, and angiogenic biomarkers such as soluble fms-like tyrosine kinase-1 (sFlt-1) and placental growth factor (PlGF) provide complementary information related to maternal health, placental perfusion, and angiogenic balance.

However, developing a clinically relevant prediction system needs more than achieving high predictive performance. The system should provide reliable predictions, interpretable explanations, and information about prediction uncertainty. It should also investigate whether interactions between clinical variables, Doppler measurements, and biomarkers provide additional predictive information.

This project proposes the development of an integrated AI framework addressing these aspects.

2 General Objective

The main objective is to design, implement, and evaluate an explainable and reliable machine learning framework for preeclampsia prediction and pregnancy outcome assessment.

The project will combine:

- multimodal clinical data;
- feature engineering;
- machine learning;
- feature selection;
- class-imbalance handling;
- prediction calibration;
- selective prediction;
- Explainable Artificial Intelligence;
- feature interaction analysis;

- pregnancy-outcome prediction;
- and software development.

The final objective is to develop a functional prototype capable of producing risk predictions together with confidence information and interpretable explanations.

3 Specific Objectives

3.1 Clinical Data Analysis

The student will study and characterize the available pregnancy dataset, including:

- maternal characteristics;
- clinical variables;
- uterine artery Doppler measurements;
- angiogenic biomarkers;
- pregnancy outcomes.

The analysis will include variable distributions, missing observations, class imbalance, correlations, and potential data-quality problems.

3.2 Multimodal Feature Representation

The student will develop a representation combining several sources of information:

$$X = [X_{\text{clinical}}, X_{\text{Doppler}}, X_{\text{biomarkers}}, X_{\text{engineered}}].$$

The contribution from each information group will be evaluated experimentally.

3.3 Feature Engineering

The project will investigate clinically motivated variables derived from the original measurements.

For example:

$$MeanPI = \frac{PI_L + PI_R}{2},$$

$$MeanRI = \frac{RI_L + RI_R}{2},$$

$$MeanPSV = \frac{PSV_L + PSV_R}{2},$$

and

$$L_{AP} = \log\left(\frac{sFlt1}{PlGF}\right).$$

The student will also investigate interaction-based representations involving Doppler and angiogenic measurements.

4 Machine Learning Development

Several machine learning models will be implemented and compared, including:

- Logistic Regression;
- Decision Tree;
- Support Vector Machine;
- K-Nearest Neighbors;
- Random Forest;
- Extra Trees;
- Gradient Boosting;
- XGBoost;
- and, where appropriate, neural-network models.

The comparison will consider:

- ROC-AUC;
- PR-AUC;
- F1-score;
- precision;
- recall;
- specificity;
- balanced accuracy;
- and calibration measures.

Repeated cross-validation will be used to obtain robust performance estimates.

5 Feature Selection and Ablation Study

A systematic feature-selection study will be conducted using approaches such as:

- variance-based filtering;
- correlation analysis;
- ANOVA;
- mutual information;
- chi-square;
- Random Forest importance;
- L1-based selection;
- SHAP-based selection.

An ablation study will progressively evaluate:

$$\textit{Clinical} \to \textit{Clinical+Doppler} \to \textit{Clinical+Doppler+Biomarkers} \to \textit{Clinical+Doppler+Biomarkers+Engineered}.$$

This study will determine the contribution from each information source.

6 Reliable Prediction

A major component will investigate prediction reliability.

The student will study model confidence and identify cases for which an automatic prediction may be inappropriate.

A selective prediction mechanism can be defined as:

$$\text{Accept prediction if } C(x) \geq \tau,$$

where $C(x)$ represents prediction confidence and τ represents a confidence threshold.

The relationship between coverage, prediction quality, confidence, and selective risk will be investigated.

The resulting system may distinguish between:

High-confidence prediction

and

Prediction needs additional clinical assessment.

7 Probability Calibration

The student will investigate whether predicted probabilities adequately reflect observed outcomes.

The study may include:

- calibration curves;
- Brier score;
- calibration error;
- Platt scaling;
- isotonic regression.

8 Explainable Artificial Intelligence

Explainability will constitute a central component of the project.

The student will use SHAP to investigate model decisions at different levels.

8.1 Global Explanations

Identify the most influential predictors across the dataset.

8.2 Local Explanations

Explain individual predictions by determining which variables contribute positively or negatively to a particular prediction.

8.3 Feature Interaction Analysis

The student will investigate pairwise SHAP interaction values:

$$\phi_{ij},$$

where ϕ_{ij} represents the interaction contribution associated with features i and j.

Potential relationships include interactions between:

- BMI and sFlt-1;
- uterine artery PI and angiogenic variables;
- sFlt-1 and PlGF;
- Doppler measurements and engineered features.

9 Explanation Stability

An additional research component will investigate whether model explanations remain stable across different training samples.

For feature j, the student may calculate its frequency among the top k predictors across cross-validation folds:

$$S_j = \frac{1}{K} \sum_{k=1}^{K} I(j \in TopK_k).$$

The objective is to determine whether important predictors remain consistent across different training partitions.

10 Counterfactual Explanations

The project may investigate counterfactual explanations.

Instead of only answering:

"Why was this patient classified as high risk?"

the system may investigate:

"Which hypothetical changes in the input profile would lead to a different model prediction?"

Counterfactual results will be treated as model-based hypothetical scenarios and not as clinical recommendations.

11 Pregnancy Outcome Prediction

The project will extend beyond preeclampsia classification.

One important regression task is the prediction of gestational age at delivery.

Possible models include:

- Linear Regression;
- Ridge Regression;
- Random Forest Regression;
- Gradient Boosting Regression;
- XGBoost Regression.

Performance will be assessed using:

$$MAE, \qquad RMSE, \qquad R^2.$$

Where the available data support it, additional pregnancy outcomes may also be investigated.

12 Multi-Outcome Prediction

As an advanced component, the student may investigate a framework capable of predicting several pregnancy outcomes from a common feature representation:

$$X \rightarrow \begin{cases} \text{Preeclampsia risk} \\ \text{Gestational age at delivery} \\ \text{Additional pregnancy outcome} \end{cases}$$

A multi-task learning approach may be compared with independent prediction models.

13 Robustness Analysis

The student will investigate model robustness under different experimental conditions, including:

- repeated cross-validation;
- bootstrap evaluation;
- different feature subsets;
- missing-data scenarios;
- perturbation of input measurements;
- different class-balancing strategies;
- different confidence thresholds.

The objective is to determine whether model performance and explanations remain stable when experimental conditions change.

14 Software Development

The Engineering component will include the development of a functional prototype.

The proposed processing pipeline is:

Patient Data → Data Validation → Preprocessing → Feature Engineering → ML Prediction → Confidence Assessment → SHAP Explanation → Clinical Report

The prototype may provide:

- patient-data entry;
- automatic preprocessing;
- preeclampsia risk prediction;
- prediction probability;
- confidence score;
- selective prediction;
- SHAP explanation;
- important feature interactions;
- gestational-age prediction;
- graphical presentation of results.

15 Engineering Project Component

For an Engineering student, the main emphasis will be placed on:

1. Software architecture;
2. data-processing pipeline;
3. machine learning implementation;
4. feature-engineering module;
5. explainability module;
6. selective-prediction module;
7. web-based interface;
8. testing and validation;
9. prototype deployment;
10. technical documentation.

The expected deliverable is a complete prototype implementing:

$$\boxed{\text{Data} \rightarrow \text{Prediction} \rightarrow \text{Confidence} \rightarrow \text{Explanation} \rightarrow \text{Clinical Report}}$$

16 Master Project Component

For a Master's student, the project will include a stronger research component.

The student will investigate:

1. multimodal feature representation;
2. feature-selection strategies;
3. interaction-based feature engineering;
4. class-imbalance strategies;
5. model calibration;
6. selective prediction;
7. SHAP interaction analysis;
8. explanation stability;
9. counterfactual explanations;
10. pregnancy-outcome prediction;
11. robustness evaluation;
12. potential multi-task learning.

The student will formulate research questions and conduct systematic experimental comparisons.

17 Expected Scientific Contributions

The project is expected to produce:

- a reproducible preprocessing pipeline for pregnancy data;
- a multimodal feature representation;
- clinically motivated engineered features;
- comparative evaluation of machine learning models;
- a reliability-aware prediction mechanism;
- global and local explainability;
- feature-interaction analysis;
- explanation-stability assessment;
- additional pregnancy-outcome models;
- and a prototype clinical decision-support system.

Depending on the experimental results, the work may provide material for scientific publications in biomedical informatics, medical AI, or healthcare computing.

18 Expected Student Skills

The project will provide practical experience in:

Artificial Intelligence

- supervised learning;
- ensemble learning;
- feature engineering;
- feature selection;
- model evaluation.

Explainable AI

- SHAP;
- global and local explanations;
- interaction analysis;
- explanation stability;
- counterfactual reasoning.

Biomedical Data Science

- clinical data preprocessing;
- multimodal data analysis;
- biomedical variables;
- pregnancy-outcome prediction.

Software Engineering

- Python;
- machine learning pipelines;
- API development;
- web application development;
- software testing;
- deployment.

Scientific Research

- experimental design;
- statistical evaluation;
- literature review;
- scientific writing;
- reproducibility.

19 Proposed Work Plan

Step	Main Activities
Step 1	Literature review and study of preeclampsia
Step 2	Dataset analysis and preprocessing
Step 3	Feature engineering and feature selection
Step 4	Baseline machine learning models
Step 5	Class imbalance and model optimization
Step 6	Calibration and selective prediction
Step 7	SHAP global and local explanations
Step 8	SHAP interaction and explanation stability
Step 9	Pregnancy-outcome prediction
Step 10	Robustness and comparative experiments
Step 11	Software prototype development
Step 12	Evaluation, documentation, and thesis writing

20 Expected Final Result

At the end of the project, the student should deliver an AI-based decision support prototype capable of receiving maternal clinical, Doppler, and angiogenic biomarker information and producing:

Preclampsia Risk
↓
Prediction Probability
↓
Confidence Assessment
↓
Selective Prediction
↓
SHAP-Based Explanation
↓
Important Feature Interactions
↓
Pregnancy-Outcome Prediction

The project combines Artificial Intelligence, biomedical informatics, explainable AI, reliable prediction, and software engineering, making it suitable for an Engineering or Master's project at ESI-SBA.

Contact

Students interested in this project are invited to contact:

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